

COOL TEMPERATE EASTERN MARGIN LAURENTIAN

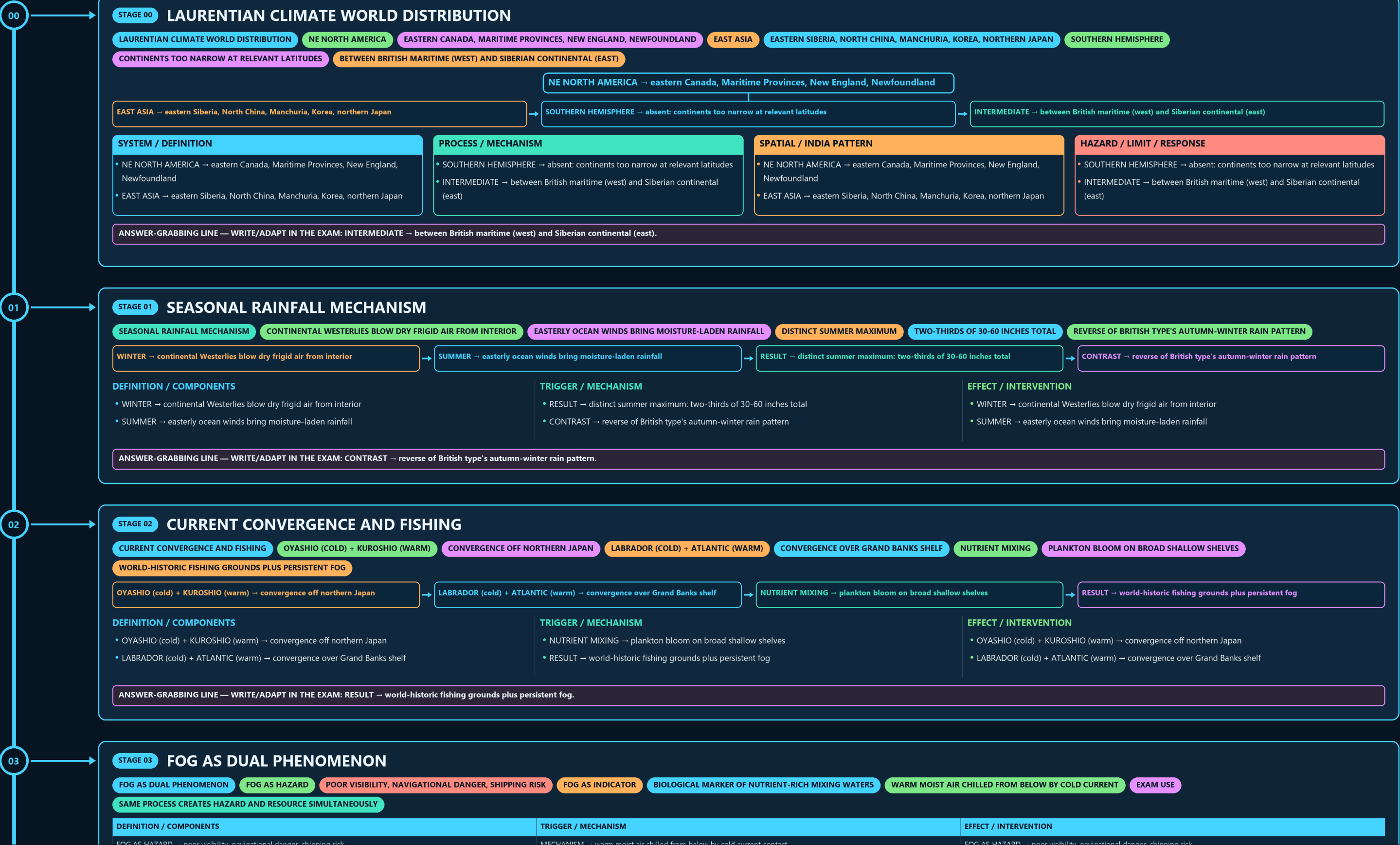
LAURENTIAN CLIMATE WORLD DISTRIBUTION → SEASONAL RAINFALL MECHANISM → CURRENT CONVERGENCE AND FISHING → FOG AS DUAL PHENOMENON → MIXED FOREST TRANSITION STRIP → FISHERY GOVERNANCE LESSON

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



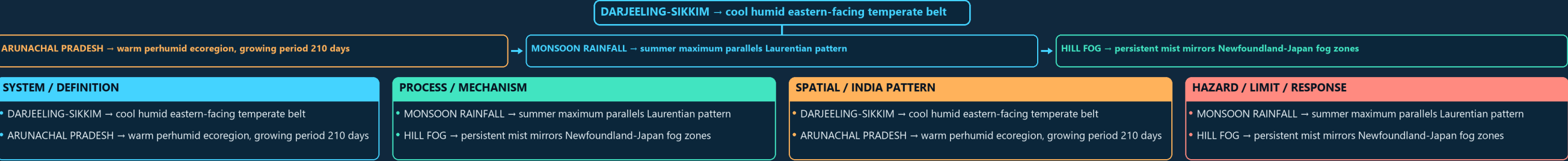
- OYASHIO (cold) + KURUSHIO (warm) → convergence off northern Japan
- LABRADOR (cold) + ATLANTIC (warm) → convergence over Grand Banks shelf

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OVEREXPLOITATION → Grand Banks cod stocks collapsed despite same oceanography.

06

STAGE 06 EASTERN HIMALAYA ANALOGUE MAP

EASTERN HIMALAYA ANALOGUE MAP COOL HUMID EASTERN-FACING TEMPERATE BELT ARUNACHAL PRADESH WARM PERHUMID ECOREGION, GROWING PERIOD 210 DAYS MONSOON RAINFALL SUMMER MAXIMUM PARALLELS LAURENTIAN PATTERN HILL FOG PERSISTENT MIST MIRRORS NEWFOUNDLAND-JAPAN FOG ZONES



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HILL FOG → persistent mist mirrors Newfoundland-Japan fog zones.

07

STAGE 07 TEESTA BASIN PROFILE

TEESTA BASIN PROFILE STEEP RELIEF RAPID RUNOFF, HIGH SEDIMENT LOAD EXTREME MONSOON RAIN TRIGGERS LANDSLIDES AND GLOF EVENTS CORRIDOR CONCENTRATION NH-10, HYDROPOWER ON NARROW VALLEY FLOOR 2023 SOUTH LHONAK GLOF



DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- STEEP RELIEF → rapid runoff, high sediment load - EXTREME MONSOON RAIN → triggers landslides and GLOF events	- CORRIDOR CONCENTRATION → NH-10, hydropower on narrow valley floor 2023 SOUTH LHONAK GLOF → illustrates combined relief-rain-infrastructure risk	- STEEP RELIEF → rapid runoff, high sediment load - EXTREME MONSOON RAIN → triggers landslides and GLOF events

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2023 SOUTH LHONAK GLOF → illustrates combined relief-rain-infrastructure risk.

08

STAGE 08 DARJEELING TEA ECONOMY

DARJEELING TEA ECONOMY DARJEELING HILLS LOW-YIELD, PREMIUM, GI-PROTECTED ORTHODOX TEA BRAHMAPUTRA VALLEY BULK-TEA BELT, HIGH-VOLUME PRODUCTION ESTATES AT 900-1800 M; TOO COLD ABOVE, TOO WARM HILL QUALITY VERSUS VALLEY QUANTITY MIRRORS ALTITUDE CONTROL

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
DARJEELING HILLS → low-yield, premium, GI-protected orthodox tea	ELEVATION → estates at 900-1800 m; too cold above, too warm below	DARJEELING HILLS → low-yield, premium, GI-protected orthodox tea
BRAHMAPUTRA VALLEY → bulk-tea belt, high-volume production	DISTINCTION → hill quality versus valley quantity mirrors altitude control	BRAHMAPUTRA VALLEY → bulk-tea belt, high-volume production

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- DARJEELING HILLS → low-yield, premium, GI-protected orthodox tea - BRAHMAPUTRA VALLEY → bulk-tea belt, high-volume production	- ELEVATION → estates at 900-1800 m; too cold above, too warm below - DISTINCTION → hill quality versus valley quantity mirrors altitude control	- DARJEELING HILLS → low-yield, premium, GI-protected orthodox tea - BRAHMAPUTRA VALLEY → bulk-tea belt, high-volume production

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DISTINCTION → hill quality versus valley quantity mirrors altitude control.

09

STAGE 09 BRITISH-LAURENTIAN COMPARISON

BRITISH-LAURENTIAN COMPARISON AUTUMN-WINTER MAX SUMMER MAX MILD, WET (MARITIME) COLD, DRY (CONTINENTAL) LAND-BASED AGRICULTURE SEA-BASED FISHING DECIDUOUS BROADLEAF

RAINFALL → British: autumn-winter max	Laurentian: summer max
WINTER → British: mild, wet (maritime)	Laurentian: cold, dry (continental)
ECONOMY → British: land-based agriculture	Laurentian: sea-based fishing
FOREST → British: deciduous broadleaf	Laurentian: mixed transitional

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- RAINFALL → British: autumn-winter max - Laurentian: summer max	- Laurentian: cold, dry (continental) - ECONOMY → British: land-based agriculture	- FOREST → British: deciduous broadleaf - Laurentian: mixed transitional

09

STAGE 09

BRITISH-LAURENTIAN COMPARISON

BRITISH-LAURENTIAN COMPARISON

AUTUMN-WINTER MAX

SUMMER MAX

MILD, WET (MARITIME)

COLD, DRY (CONTINENTAL)

LAND-BASED AGRICULTURE

SEA-BASED FISHING

DECIDUOUS BROADLEAF

RAINFALL → British: autumn-winter max

WINTER → British: mild, wet (maritime)

ECONOMY → British: land-based agriculture

FOREST → British: deciduous broadleaf

Laurentian: summer max

Laurentian: cold, dry (continental)

Laurentian: sea-based fishing

Laurentian: mixed transitional

DEFINITION / COMPONENTS

- RAINFALL → British: autumn-winter max
- Laurentian: summer max
- WINTER → British: mild, wet (maritime)

TRIGGER / MECHANISM

- Laurentian: cold, dry (continental)
- ECONOMY → British: land-based agriculture
- Laurentian: sea-based fishing

EFFECT / INTERVENTION

- FOREST → British: deciduous broadleaf
- Laurentian: mixed transitional

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FOREST → British: deciduous broadleaf | Laurentian: mixed transitional.

10

STAGE 10

POLAR-VORTEX HYPOTHESIS

POLAR-VORTEX HYPOTHESIS

STRATOSPHERIC DISRUPTION

WEAKENS POLAR VORTEX CONTAINMENT

COLD ARCTIC AIR

SPILLS SOUTHWARD INTO MID-LATITUDES

LAURENTIAN REGIONS

EXPERIENCE SEVERE COLD OUTBREAKS

ACTIVELY DEBATED LINK TO ARCTIC SEA-ICE LOSS; NOT SETTLED

STRATOSPHERIC DISRUPTION → weakens polar vortex containment

COLD ARCTIC AIR → spills southward into mid-latitudes

LAURENTIAN REGIONS → experience severe cold outbreaks

STATUS → actively debated link to Arctic sea-ice loss; not settled science

DEFINITION / COMPONENTS

- STRATOSPHERIC DISRUPTION → weakens polar vortex containment
- COLD ARCTIC AIR → spills southward into mid-latitudes

TRIGGER / MECHANISM

- LAURENTIAN REGIONS → experience severe cold outbreaks
- STATUS → actively debated link to Arctic sea-ice loss; not settled science

EFFECT / INTERVENTION

- STRATOSPHERIC DISRUPTION → weakens polar vortex containment
- COLD ARCTIC AIR → spills southward into mid-latitudes

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STATUS → actively debated link to Arctic sea-ice loss; not settled science.

11

STAGE 11

LAURENTIAN ANSWER SPINE

LAURENTIAN ANSWER SPINE

INTERMEDIATE TYPE BETWEEN BRITISH MARITIME AND SIBERIAN CONTINENTAL

NE NORTH AMERICA AND EAST ASIA ONLY; ABSENT IN

COLD DRY WINTERS, SUMMER RAIN MAXIMUM, CURRENT CONVERGENCE, FOG-FISHERY

INDIA EASTERN HIMALAYA ANALOGUE; TRANSPARENT ZERO-DIRECT-PYQ BOUNDARY

DEFINE → intermediate type between British maritime and Siberian continental

LOCATE → NE North America and East Asia only; absent in Southern Hemisphere

EXPLAIN → cold dry winters, summer rain maximum, current convergence, fog-fishery

QUALIFY → India Eastern Himalaya analogue; transparent zero-direct-PYQ boundary

DEFINITION / COMPONENTS

- DEFINE → intermediate type between British maritime and Siberian continental
- LOCATE → NE North America and East Asia only; absent in Southern Hemisphere

TRIGGER / MECHANISM

- EXPLAIN → cold dry winters, summer rain maximum, current convergence, fog-fishery
- QUALIFY → India Eastern Himalaya analogue; transparent zero-direct-PYQ boundary

EFFECT / INTERVENTION

- DEFINE → intermediate type between British maritime and Siberian continental
- LOCATE → NE North America and East Asia only; absent in Southern Hemisphere

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → India Eastern Himalaya analogue; transparent zero-direct-PYQ boundary.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

EASTERN HIMALAYAS (KHULLAR)

SIKKIM + ARUNACHAL + HILLY ASSAM; WARM PERHUMID

SIKKIM IS DRAINED BY THE TEESTA SYSTEM; TEESTA ULTIMATELY

DARJEELING TEA

HUMID HILL CLIMATE AND MIDDLE-ELEVATION ESTATES; EXACT RAIN/ELEVATION/YIELD RANGES

SUMMER (MONSOON) RAINFALL MAXIMUM + HILL FOG — THE

THE TEESTA IS A TRIBUTARY OF THE GANGA

OPTIONAL PROCESS DEPTH

- Eastern Himalayas (Khullar) = Sikkim + Arunachal + hilly Assam; warm perhumid, growing period 210 days.
- Sikkim is drained by the Teesta system; Teesta ultimately joins the Brahmaputra/Jamuna.
- Darjeeling tea: humid hill climate and middle-elevation estates; exact rain/elevation/yield ranges are source-era generalisations.
- Summer (monsoon) rainfall maximum + hill fog — the eastern-margin temperate signature.

DATA / INSTITUTION LIMIT

- NOT: The Teesta is a tributary of the Ganga → it is a tributary of the Brahmaputra.
- NOT: Darjeeling produces India's bulk tea → it is low-yield premium tea; Assam/Brahmaputra valley
- Carrying capacity of Himalayan hill regions: how tourism, hydropower and unplanned construction
- Altitude-controlled tea geography: why Darjeeling yields premium low-volume tea while Assam yields bulk tea.

CONTEMPORARY PARALLEL

- Subject: Geography · Tier: Advanced (India-applied) · GS Paper: GS-I / GS-III Grounded in: D.R.
- Khullar, India: A Comprehensive Geography + Majid Husain, India geography + verified CA.
- = from source book · CAUTION: = inference / standard knowledge · = current affairs.
- Sikkim is drained chiefly by the Teesta and its tributaries; the Teesta joins the Brahmaputra (Jamuna) system in Bangladesh.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.